

addition, no improvement in implantation rates was observed in the study group. However, significant difference ($P = 0.04$) in clinical pregnancy rate was observed with the EmbryoGlue in patients with previous IVF failure. In the study group, 50% of patients (6/12) with previous IVF failure had successful implantation, but in the control group none of the patients (0/11) with previous implantation failure could achieve pregnancy. It was concluded; it is difficult to conclude a favorable role of EmbryoGlue in IVF-ET cycles with a good prognosis. However, in patients with recurrent implantation failure, it may be considered as a useful transfer medium.

Hyaluronan binding assay (HBA)

There is insufficient evidence supporting the clinical utility of HBA testing as an advanced sperm selection technique. More studies are needed to support improved outcomes (i.e., increased successful pregnancies with delivery of liveborn children).

In a prospective study conducted by Lazarevic et. al. (2010), the question was examined: Hyaluronan binding assay (HBA) versus sperm penetration assay (SPA): Can HBA replace the SPA test in male partner screening before in vitro fertilization?

Semen samples from 26 infertility couples were analyzed. Both, normal and male factor patients were included. The data obtained in this study showed no statistically significant relationship between the HBA and SPA results. The mean HBA scores 76.3%, 61.3% and 76.8% were statistically not significantly different as compared to patients with negative (<5), grey zone (5–8) and for positive (>8) sperm capacitation index values.

It was concluded that the HBA is not predictive of the results of the SPA. Therefore, HBA test does not reduce the need for and cannot replace the SPA test in male partner screening prior to infertility treatment.

In vitro maturation of oocytes

Although preliminary results with in vitro maturation are promising, studies to date show that implantation and pregnancy rates are significantly lower than those achieved with standard IVF. Further evidence from well-designed trials is needed to determine the long-term safety and efficacy of the procedure.

In vitro maturation (IVM) is a technique used to induce immature oocytes collected in different periods of embryonic growth.

A prospective study by Carles, et. al. (2023) evaluated the competence of our in vitro maturation autologous co-culture method on the maturation and developmental potential of immature oocytes obtained from stimulated IVF-ICSI cycles, to obtain additional embryos for the couple as a rescue system to increase the changes of cumulative pregnancy. Fourteen couples managed in IVF-ICSI in our center,